

Assignment 1: **Text Classification**

Task #1: Tweet Emotions' Classification

You should use CoLab to complete this task.

Upload the dataset 'tweet_emotions.csv' (available on LMS) on your Google Drive, read it from within CoLab (10 Marks)

This dataset contains a 'Tweet' column that contains tweets from different users and a 'Label' column which indicates the emotion of tweet (joy, surprise, sadness, fear, anger, disgust).

Use appropriate pandas functions to display number of instances in each of the class labels (10 Marks)

Prepare data that you can feed to dense neural network, remember text cannot directly be injected to the network, you need to convert it in 0-1 values. Use tokenizer and other classes to prepare the data. (20 Marks)

Use 80% of each of the class labels to train a dense network with appropriate input and output layers and two hidden layers with 64 and 32 neuron respectively (10 Marks)

Test the remaining 20% of the data, report test and training accuracy (10 Marks)

Inside your notebook, also answer following questions,

Does accuracy vary from class to class label?

If so, write the reasons

Task #2: BBC Sports News Classification

You should use CoLab to complete this task.

Download BBC Sports Dataset available on LMS

Upload it on your Google Drive, read it from within CoLab (10 Marks)

This dataset contains new stories belonging to five classes (athletics, cricket, football, rugby, tennis)

Prepare data that you can feed to dense neural network, remember text cannot directly be injected to the network, you need to convert it in 0-1 values. Use tokenizer and other classes to prepare the data. (5 Marks)

Use 80% of each of the class labels to train a dense network with appropriate input and output layers and two hidden layers with 64 and 32 neuron respectively (5 Marks)

Test the remaining 20% of the data, report test and training accuracy (5 Marks)

Inside your note book, also answer following questions, (5 Marks)

Does accuracy vary from class to class label?

If so, write the reasons

Submission instructions: once you are done, please download the ipynb file and submit to this assignment. Please download and upload the file rather sharing the link only.

bbcsports.csv bbcsports.csv19 January 2026, 10:19 AM

tweet_emotions.csv tweet_emotions.csv19 January 2026, 10:19 AM